

Project Overview

Project Link - https://nbviewer.org/github/npstanwar/medibuddy-user-behavior-analysis/blob/temp-fix/MediBuddy_Capstone_Project_V1.0.ipynb

This report summarizes the insights and findings from the MediBuddy Insurance Claim Prediction and Risk Analysis project.

The objective of this project was to analyze healthcare insurance data, identify the major risk drivers influencing insurance costs,

and develop a predictive machine learning model for estimating claim amounts.

Dataset Description

Two datasets were merged using policy number: personal details and medical-cost information.

Features include age, BMI, gender, number of children, region, smoking status, and insurance charges.

Total records: 1338.

Target variable: Insurance charges (INR).

Exploratory Data Analysis Summary

EDA revealed that insurance charges are heavily right-skewed, indicating high-cost outliers. Smokers incur dramatically higher claims than non-smokers.

Higher BMI and increasing age strongly increase claim amounts.

Gender and region show minimal influence on insurance cost.

Interaction analysis shows that smokers with high BMI generate extremely high costs.

Model Development

Three models were trained and evaluated:

Linear Regression as a baseline and interpretability model. Random Forest to capture nonlinear interactions.

Gradient Boosting as the final production-grade model.

Model Performance

Linear Regression: $R^2 = 0.78$, RMSE ≈ 5796 INR.

Random Forest: $R^2 = 0.86$, RMSE ≈ 4594 INR.

Gradient Boosting: $R^2 = 0.879$, RMSE ≈ 4335 INR.

Gradient Boosting was selected as the final model due to highest accuracy and stable generalization.

Feature Importance

1. Smoking status contributes approximately 68% of total importance.
2. BMI contributes about 19%.
3. Age contributes about 12%.
4. Children contributes around 1%.
5. Gender and region contribute negligibly.

This confirms that lifestyle and medical variables dominate insurance risk.

Business Question Answers

1. Gender is not a relevant pricing constraint due to negligible effect. Average policy cost is defined by the mean insurance charge. Geographic pricing is not recommended.

2. Dependents have only minor influence on cost.

3. BMI is a strong risk indicator.

Smoking status is the most critical pricing factor.

Age has a strong positive relationship with cost.

BMI-based discounts are recommended for low-risk individuals.

Recommendations

1. Apply strong smoker surcharges.
2. Introduce BMI-based premium adjustments.
3. Increase premiums gradually with age.
4. Offer discounts to non-smokers with normal BMI.
5. Avoid pricing based on gender and region.
6. Deploy Gradient Boosting for automated premium estimation.

Conclusion

The final model explains nearly 88% of claim variability and provides reliable insights for insurance pricing.

This project demonstrates how machine learning can support data-driven decision-making in healthcare insurance.